



ACCURACY AND REPEATABILITY OF HUMAN BODY MEASURING ANALYSIS

3DLOOK

KEY FIGURES

4

MEASURING DAYS

8

MEASURING STATIONS

14

COMPANIES PARTICIPATED

27

PARTICIPANTS
FROM 8 COUNTRIES

72

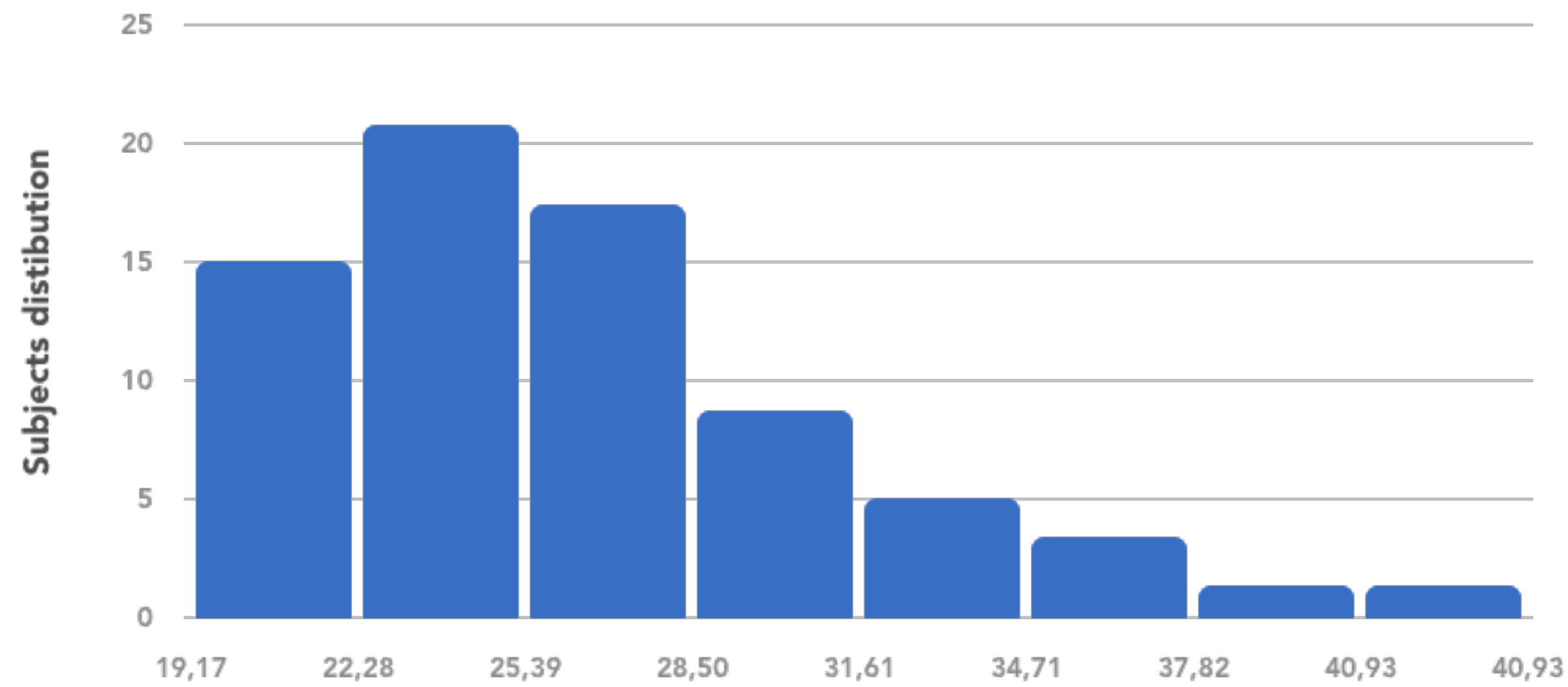
SUBJECTS MEASURED

1,152

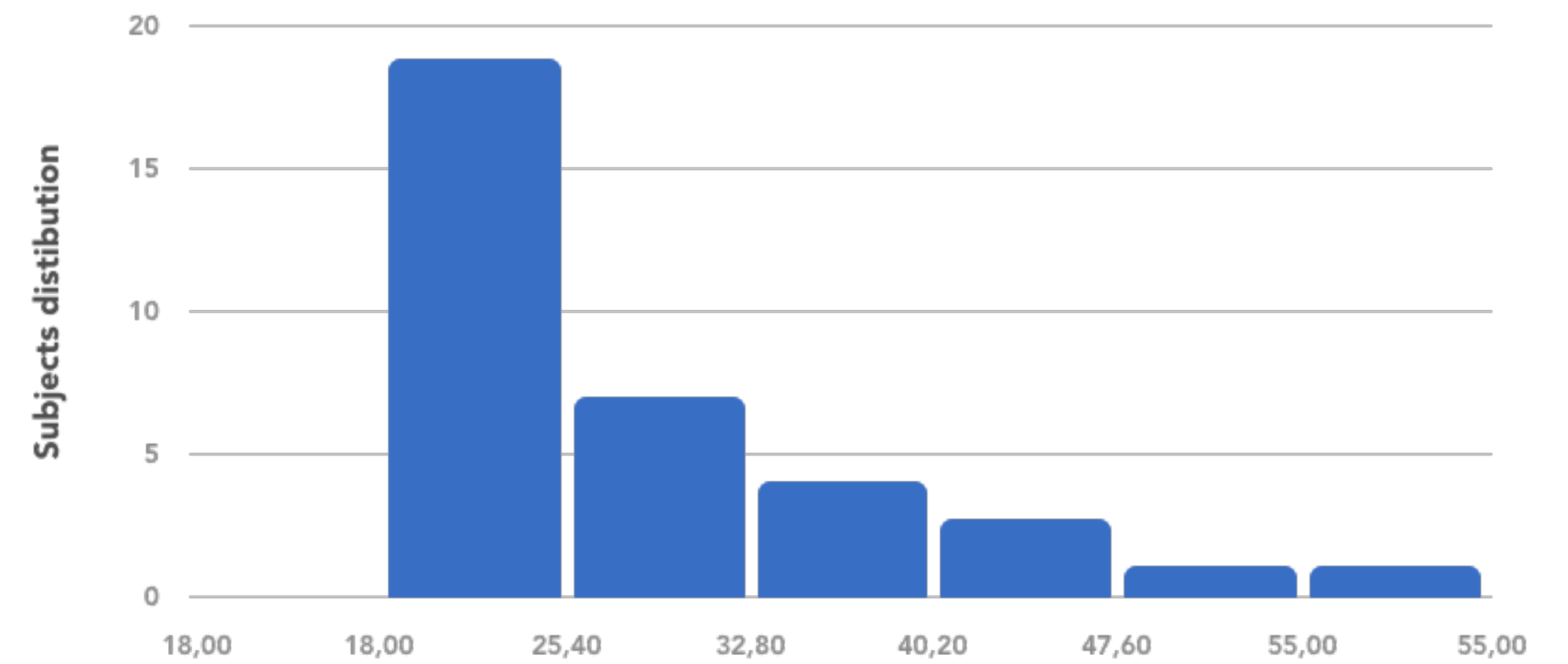
DATA POINTS GATHERED

SUBJECT SAMPLE

BMI Distribution



Age Distribution



METHODOLOGY: MEASUREMENTS

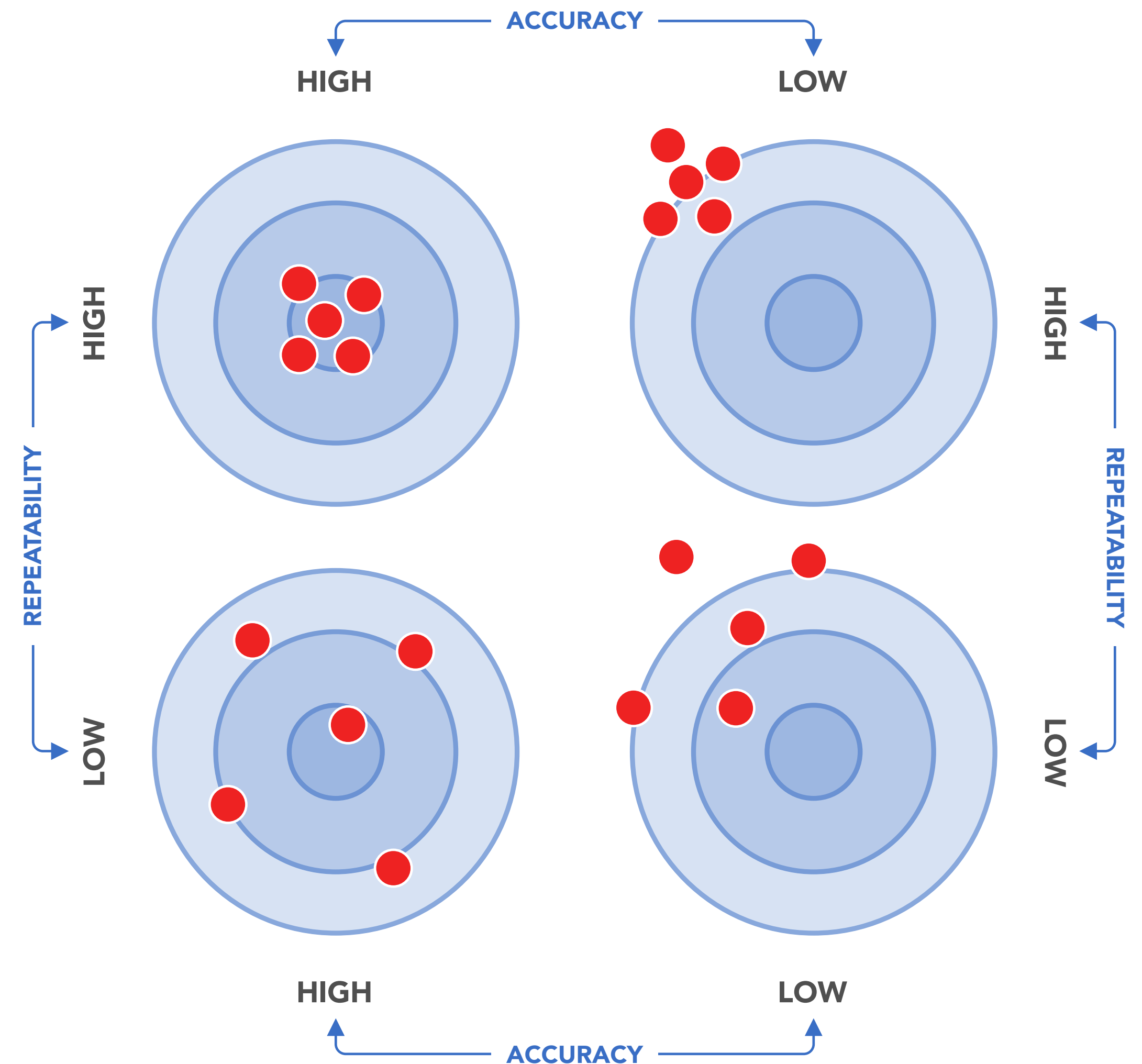
11 ISO-compatible body measurements used

Terminology	Source
Neck girth	ISO 8559-1:2017, 5.3.2
Back neck point to waist	ISO 8559-1:2017, 5.4.5
Upper arm girth (R)	ISO 8559-1:2017, 5.3.16
Back neck point to width (R)	ISO 8559-1:2017, 5.3.17
Across back shoulder width	ISO 8559-1:2017, 5.4.3
Bust girth	ISO 8559-1:2017, 5.3.4
Waist girth	ISO 8559-1:2017, 5.3.10
Hip girth	ISO 8559-1:2017, 5.3.13
Thigh girth (R)	ISO 8559-1:2017, 5.3.20
Total crotch length	ISO 8559-1:2017, 5.4.18
Inside leg height (R)	ISO 8559-1:2017, 5.1.15

UNDERSTANDING ACCURACY AND REPEATABILITY

Accuracy is a measure of the degree of closeness of a measured or calculated value to its actual value – in our case to the average measurement from 3D scanners.

Repeatability is defined as the closeness of agreement between independent test results (in our case two measuring sessions) and shows how close they are to each other.



3DLOOK

REPEATABILITY ANALYSIS

Each measurement station (experts, 3D body scanners, mobile solution – 3DLOOK) measured every participant twice.

Repeatability analysis shows the difference (in cm) between two measuring sessions.

Lower values in each measurement mean more consistent result.



**Experts – manual
measurements**



3D Scanners

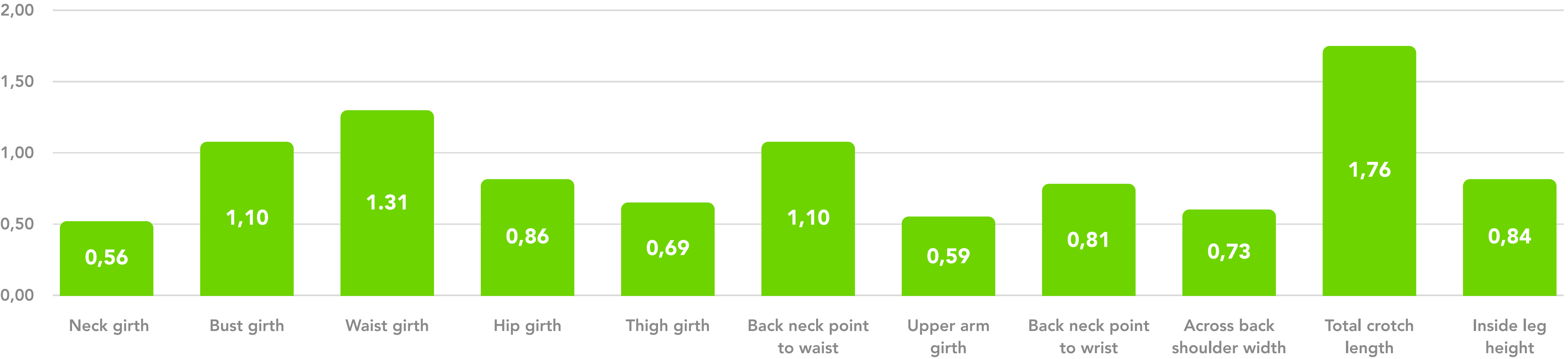


**Mobile Solution –
3DLOOK**

EXPERT MANUAL MEASURING REPEATABILITY ANALYSIS

The difference (cm) between two measuring sessions

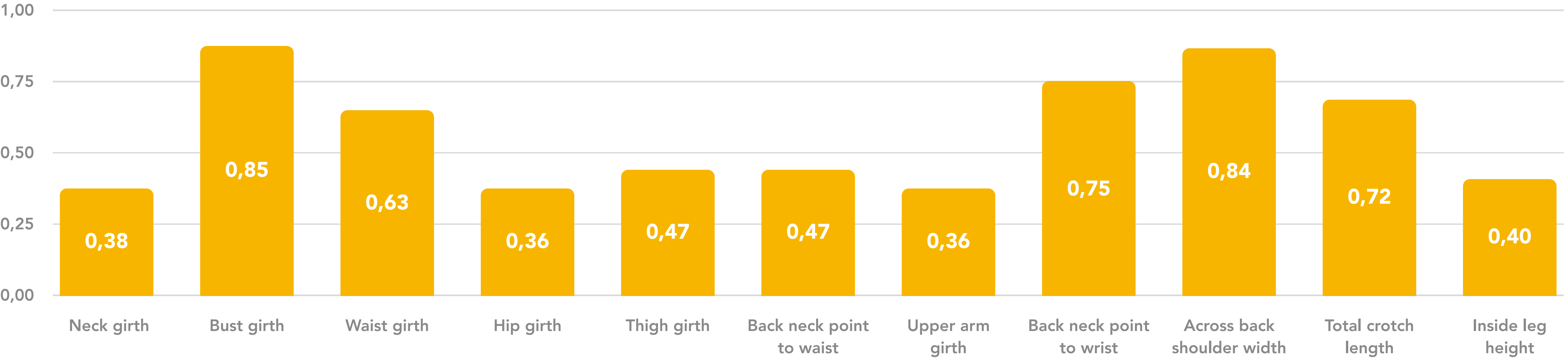
Measuring station	Neck girth	Bust girth	Waist girth	Hip girth	Thigh girth	Back neck point to waist	Upper arm girth	Back neck point to wrist	Across back shoulder width	Total crotch length	Inside leg height
Expert 1	0,52	1,25	0,91	0,88	0,78	1,05	0,82	0,94	0,65	1,78	1,07
Expert 5	0,60	0,94	1,70	0,84	0,59	1,16	0,37	0,67	0,82	1,73	0,60
AVG expert	0,56	1,10	1,31	0,86	0,69	1,10	0,59	0,81	0,73	1,76	0,84



SCANS REPEATABILITY ANALYSIS

The difference (cm) between two measuring sessions

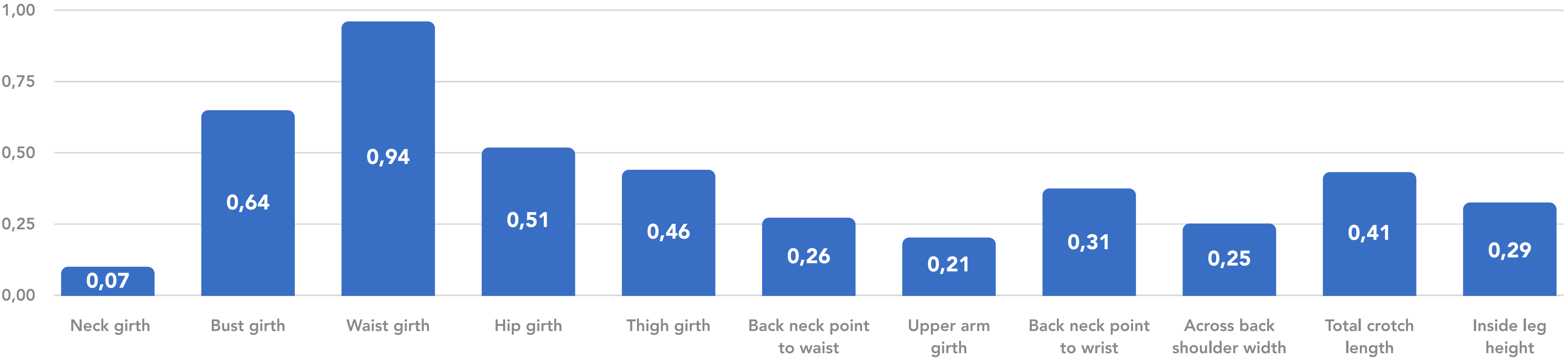
Measuring station	Neck girth	Bust girth	Waist girth	Hip girth	Thigh girth	Back neck point to waist	Upper arm girth	Back neck point to wrist	Across back shoulder width	Total crotch length	Inside leg height
3D scanner 1	0,23	0,85	0,65	0,31	0,26	0,51	0,27	0,56	0,70	0,80	0,31
3D scanner 2	0,52	0,85	0,61	0,41	0,67	0,43	0,46	0,95	0,98	0,63	0,48
AVG scan	0,38	0,85	0,63	0,36	0,47	0,47	0,36	0,75	0,84	0,72	0,40



3DLOOK REPEATABILITY ANALYSIS

The difference (cm) between two measuring sessions

Measuring station	Neck girth	Bust girth	Waist girth	Hip girth	Thigh girth	Back neck point to waist	Upper arm girth	Back neck point to wrist	Across back shoulder width	Total crotch length	Inside leg height
3DLOOK	0,07	0,64	0,94	0,51	0,46	0,26	0,21	0,31	0,25	0,41	0,29



COMPARISON OF REPEATABILITY: EXPERTS, 3D SCANNERS AND 3DLOOK

The difference (cm) between two measuring sessions

Measuring station	Neck girth	Bust girth	Waist girth	Hip girth	Thigh girth	Back neck point to waist	Upper arm girth	Back neck point to wrist	Across back shoulder width	Total crotch length	Inside leg height
3DLOOK	0,07	0,64	0,94	0,51	0,46	0,26	0,21	0,31	0,25	0,41	0,29
avg 3D scanner	0,38	0,85	0,63	0,36	0,47	0,47	0,36	0,75	0,84	0,72	0,40
AVG expert	0,56	1,10	1,31	0,86	0,69	1,10	0,59	0,81	0,73	1,76	0,84

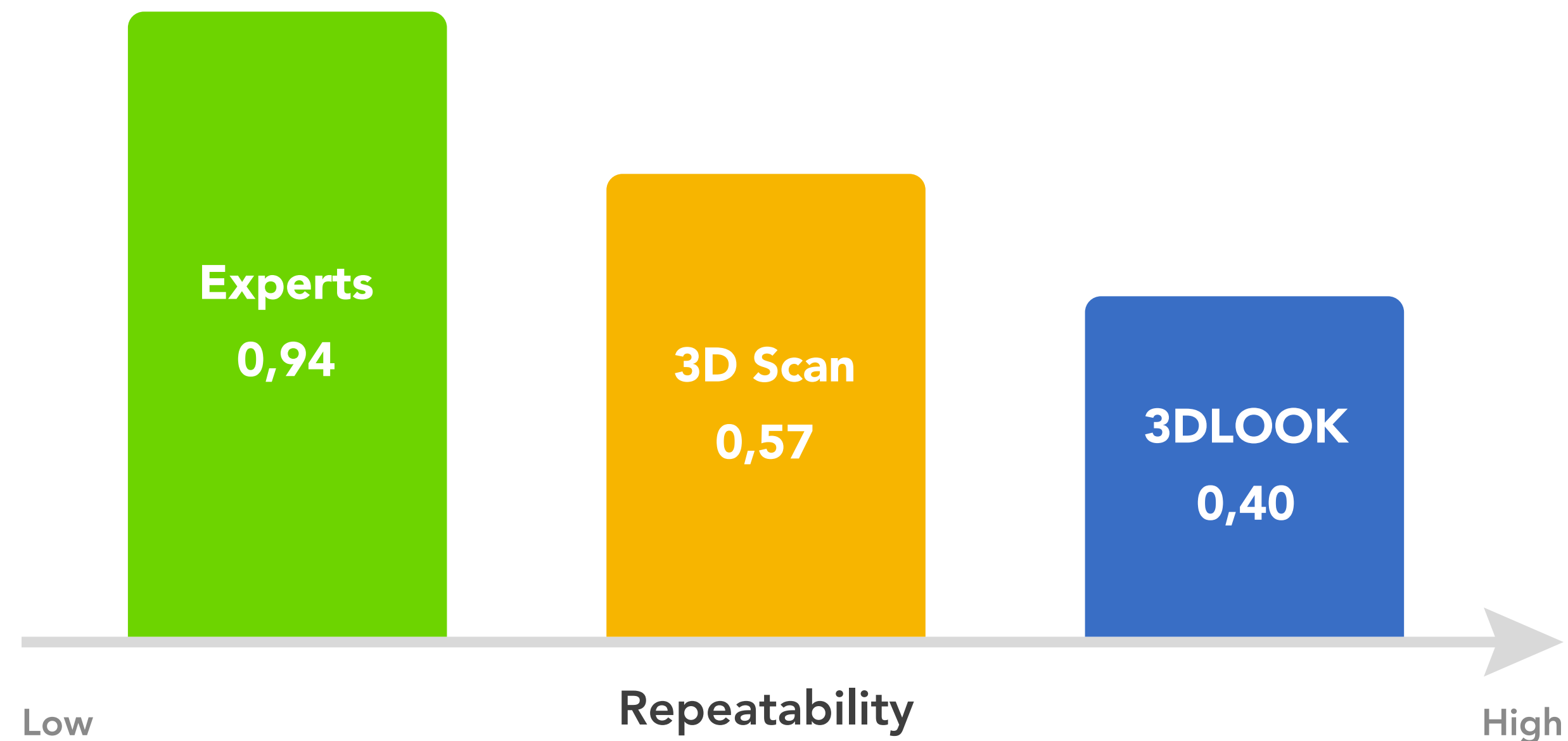


CONCLUSION

On average, 3DLOOK is better in terms of repeatability. With the lowest value of **0.07 cm**, the highest of **0.94 cm** and the average of **0.4 cm**, 3DLOOK is the most consistent among all participants.

3D scanners fall a little behind with the average of 0.57 cm.

Experts were the most inconsistent. The big discrepancy in waist and total crotch length measurements is due to the difficulty of taking these measurements manually.



3DLOOK

ACCURACY ANALYSIS. METHODOLOGY

We used the average measurements from 3D scanners as a reference.

Accuracy analysis shows the difference between manual measurements / 3DLOOK measurements and the measurements from 3D scanners.

Lower values mean higher accuracy.



**Experts – manual
measurements**

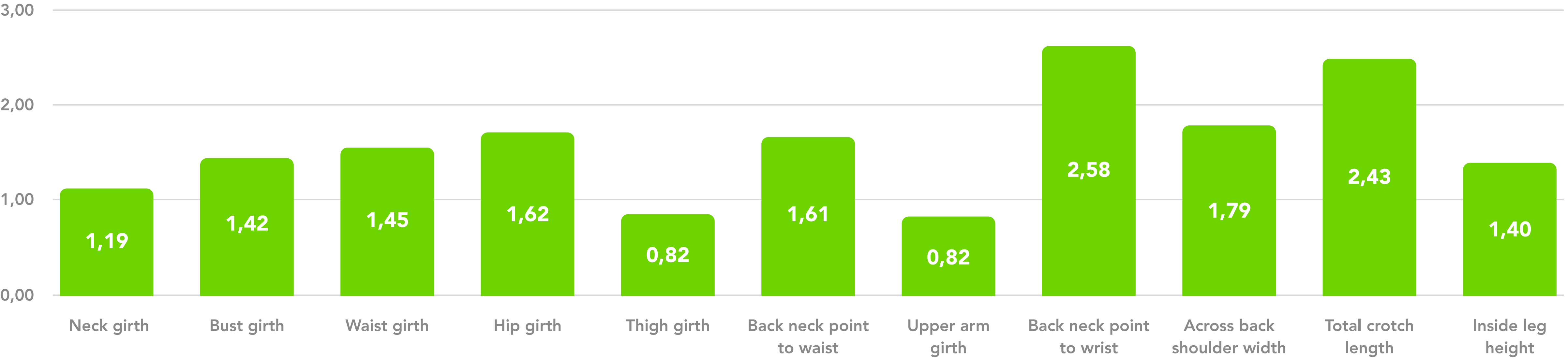


**Mobile Solution –
3DLOOK**

COMPARISON OF EXPERT MANUAL MEASUREMENTS WITH AVERAGE MEASUREMENTS FROM 3D SCANNERS

The difference (cm). The average value from 2 measuring sessions was used.

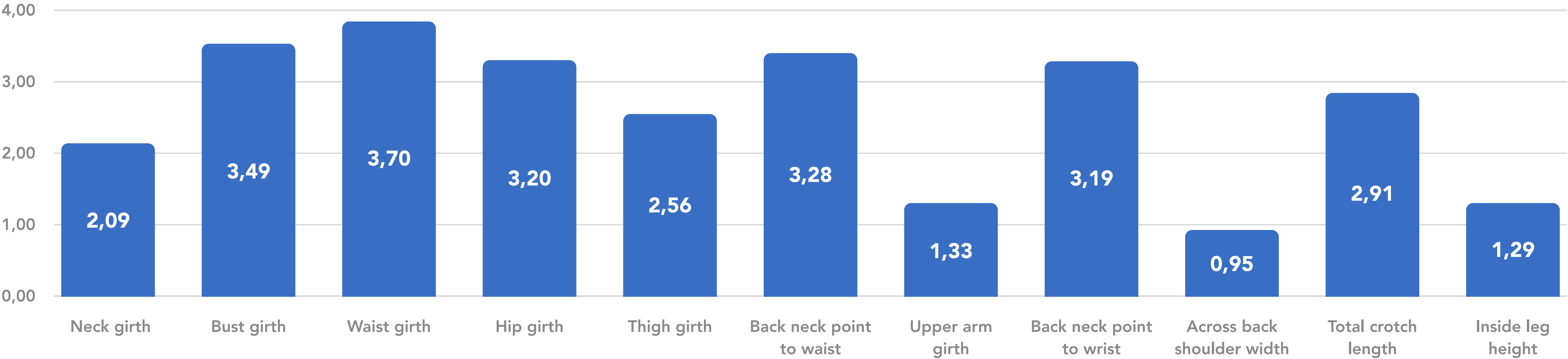
Measuring station	Neck girth	Bust girth	Waist girth	Hip girth	Thigh girth	Back neck point to waist	Upper arm girth	Back neck point to wrist	Across back shoulder width	Total crotch length	Inside leg height
AVG Expert	1,19	1,42	1,45	1,62	0,82	1,61	0,82	2,58	1,79	2,43	1,40



COMPARISON OF 3DLOOK MEASUREMENTS WITH AVERAGE MEASUREMENTS FROM 3D SCANNERS

The difference (cm). The average value from 2 measuring sessions was used.

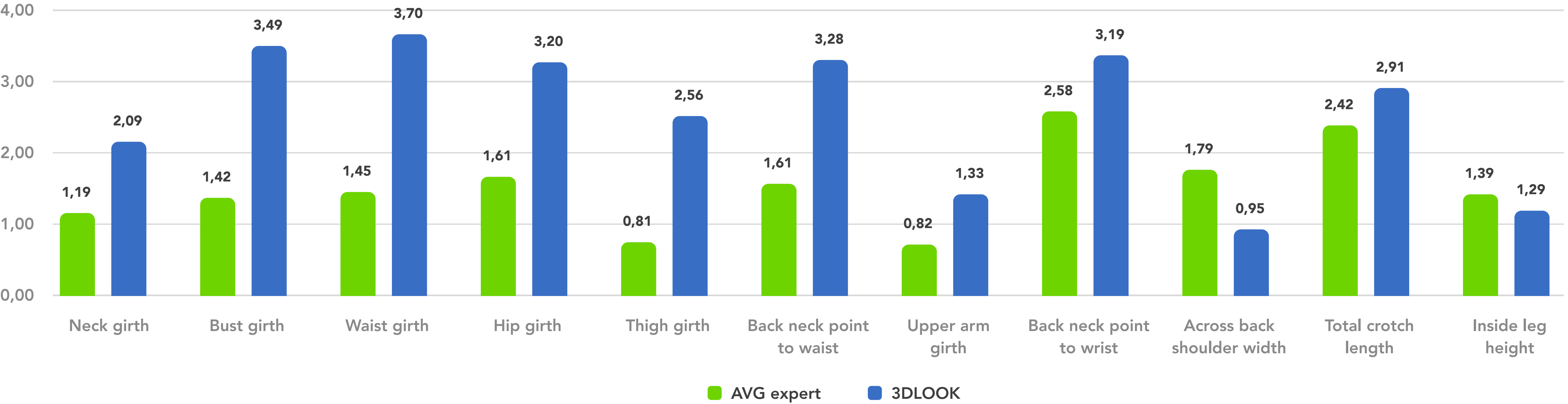
Measuring station	Neck girth	Bust girth	Waist girth	Hip girth	Thigh girth	Back neck point to waist	Upper arm girth	Back neck point to wrist	Across back shoulder width	Total crotch length	Inside leg height
3DLOOK	2,09	3,49	3,70	3,20	2,56	3,28	1,33	3,19	0,95	2,91	1,29



COMBINED ACCURACY TEST

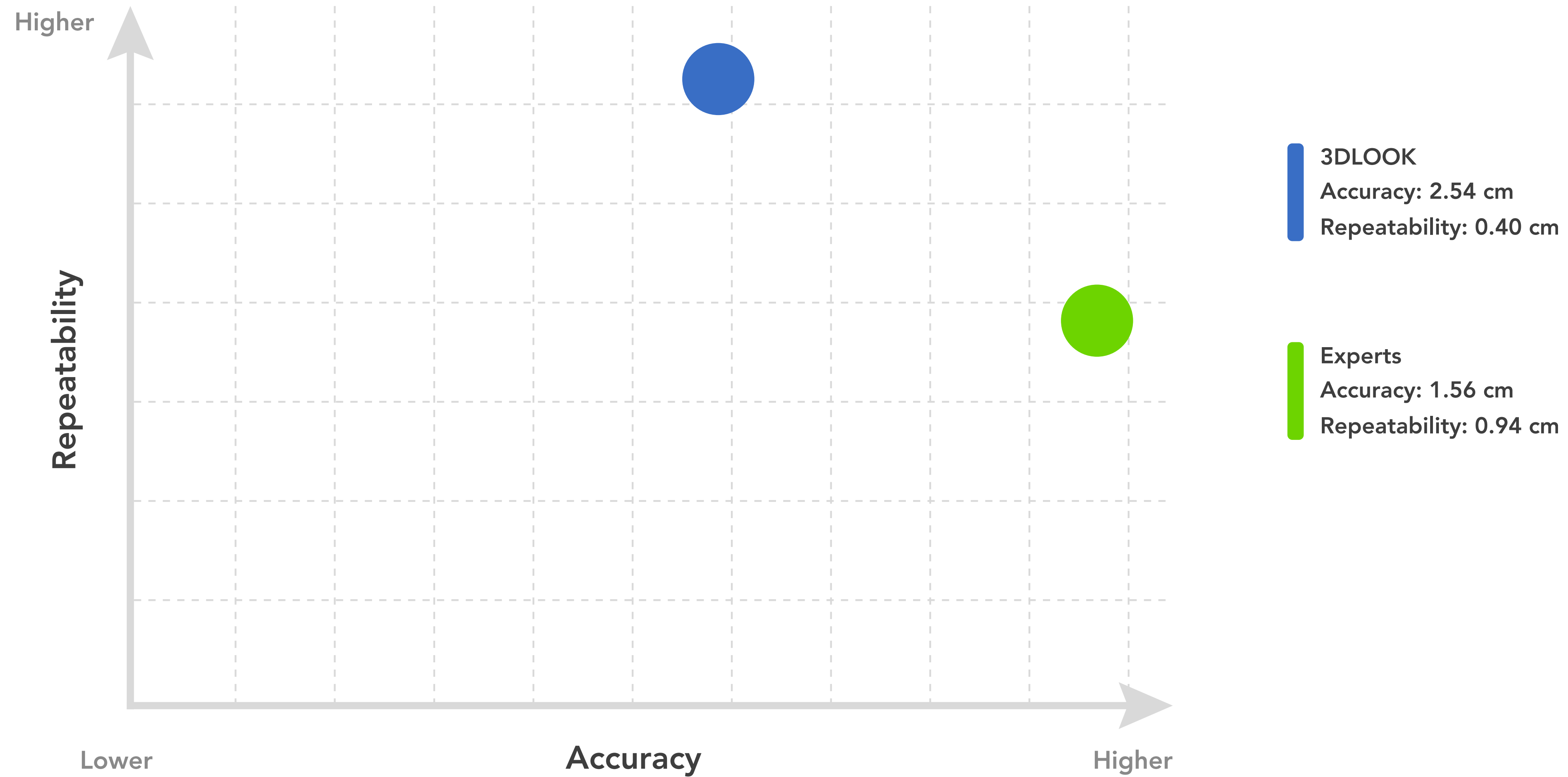
The difference (cm)

Measuring station	Neck girth	Bust girth	Waist girth	Hip girth	Thigh girth	Back neck point to waist	Upper arm girth	Back neck point to wrist	Across back shoulder width	Total crotch length	Inside leg height
AVG Expert	1,19	1,42	1,45	1,61	0,81	1,61	0,82	2,58	1,79	2,42	1,39
3DLOOK	2,09	3,49	3,70	3,20	2,56	3,28	1,33	3,19	0,95	2,91	1,29



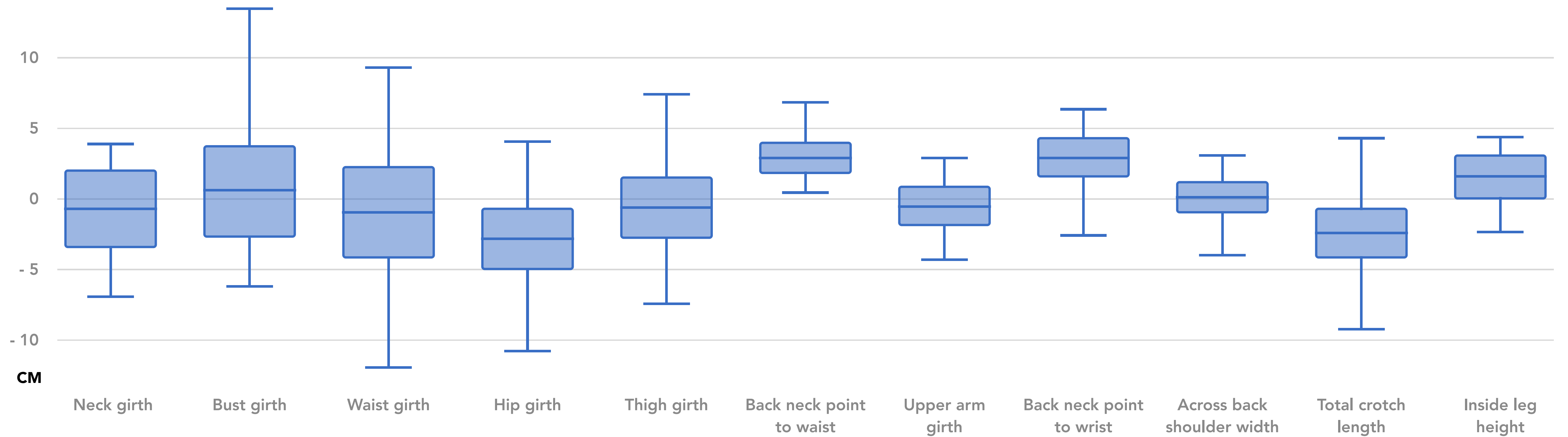
3DLOOK

CONCLUSION



3DLOOK

BOX PLOTS



3DLOOK ACCURACY COMPARED TO MEASUREMENTS FROM A 3D SCANNER

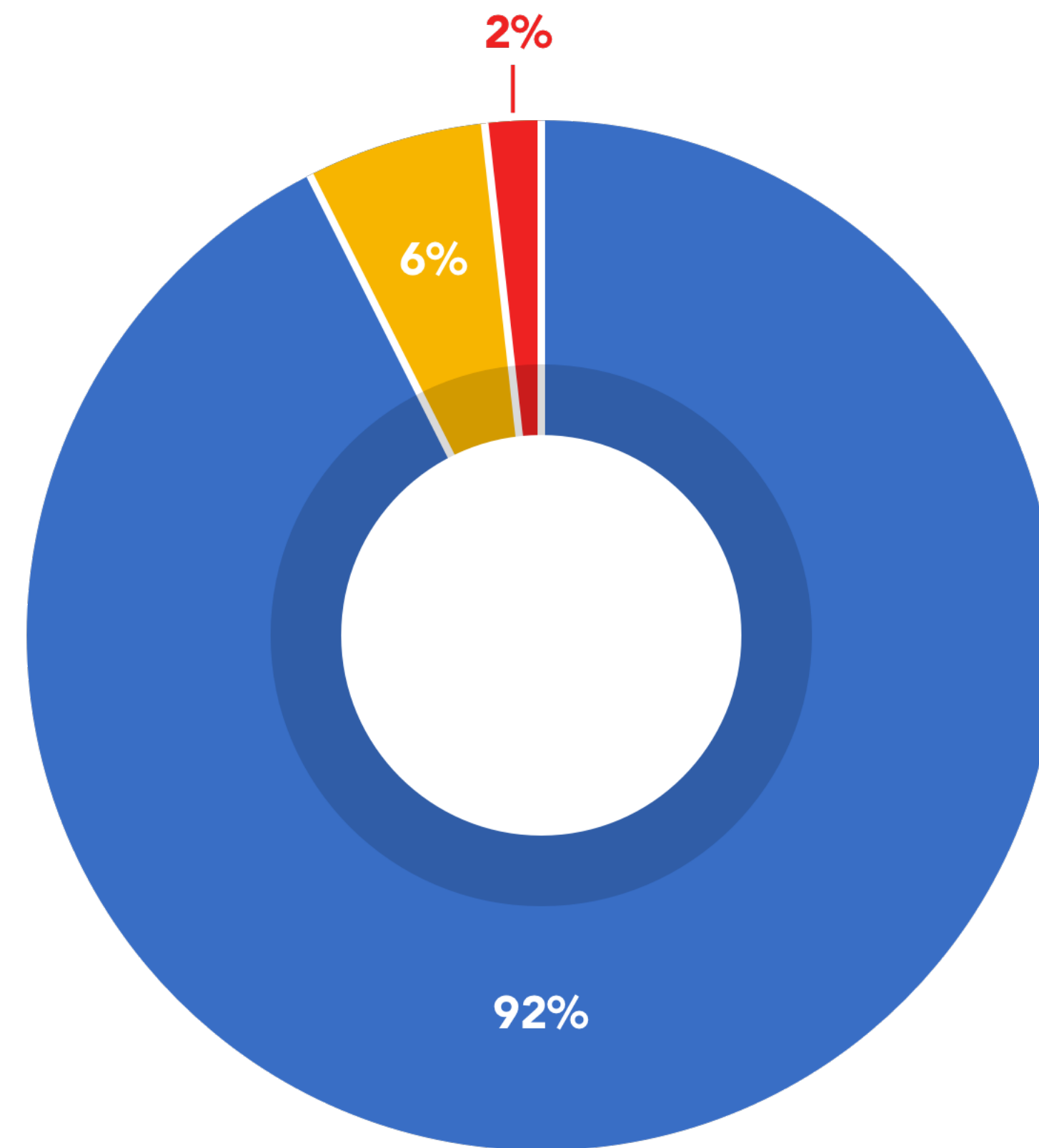
Box Plots show the distribution of discrepancy for each measurement. The best result is in a 0 area with the smallest maximum and minimum values.



CUSTOMER FEEDBACK

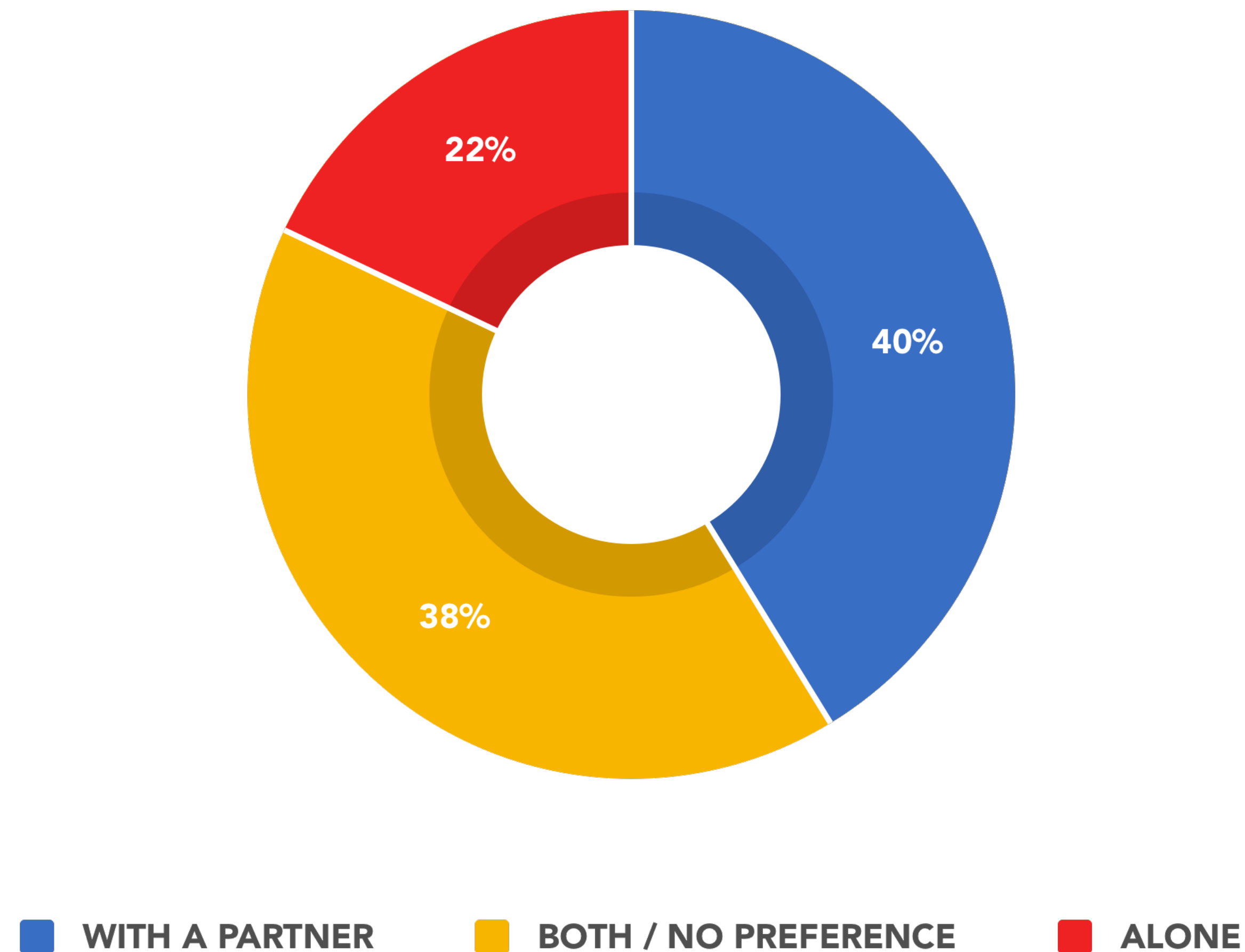
3DLOOK

SATISFACTION



■ VERY SATISFIED WITH THE EVENT ■ IT WAS NOT BAD ■ RATHER DISAPPOINTED, I WAS EXPECTING SOMETHING ELSE

USER PREFERENCE FOR TAKING PHOTOS

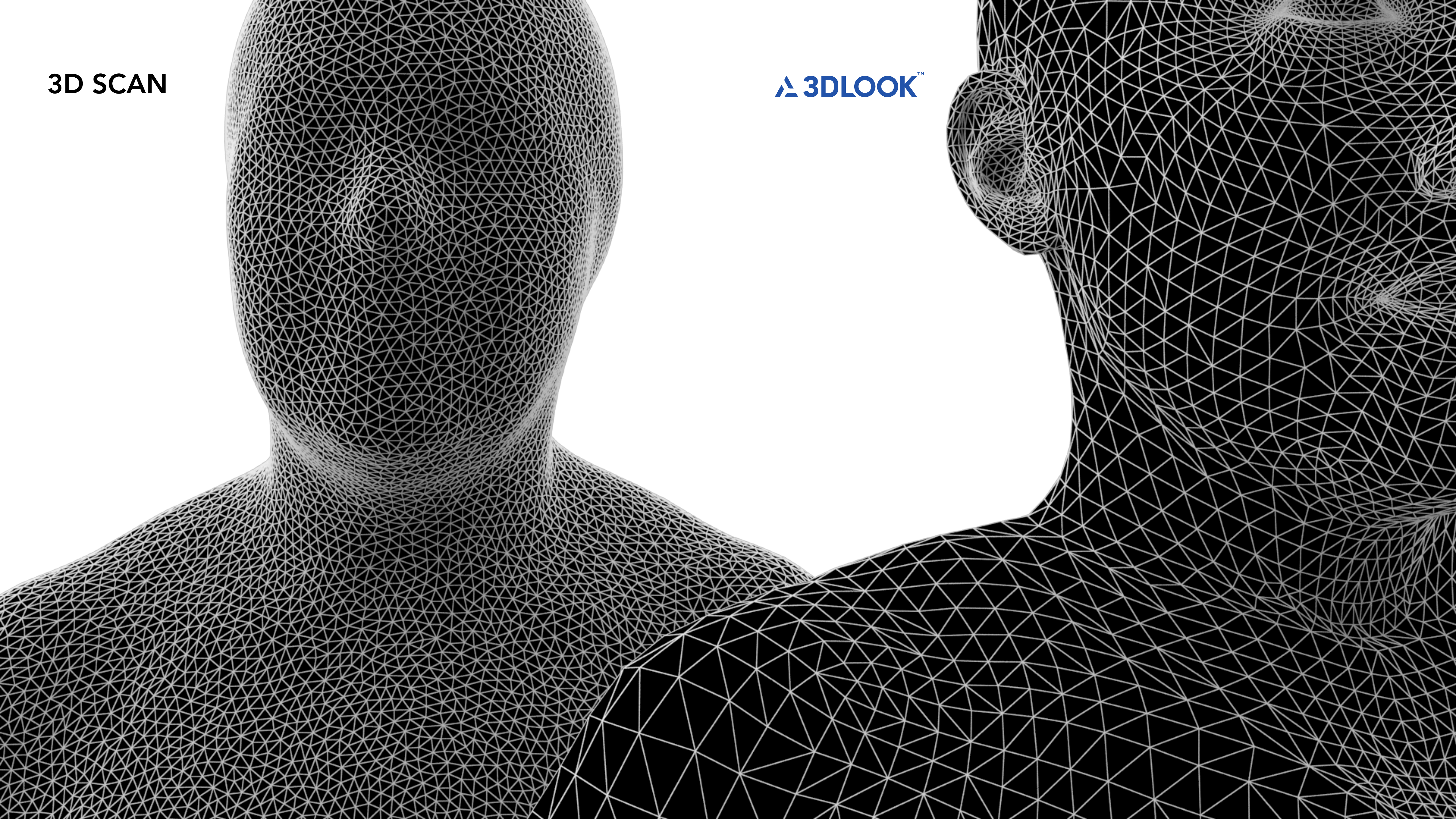




QUALITY OF 3D AVATARS

3D SCAN

 3DLOOK™



3D SCAN



3DLOOK™



3D SCAN

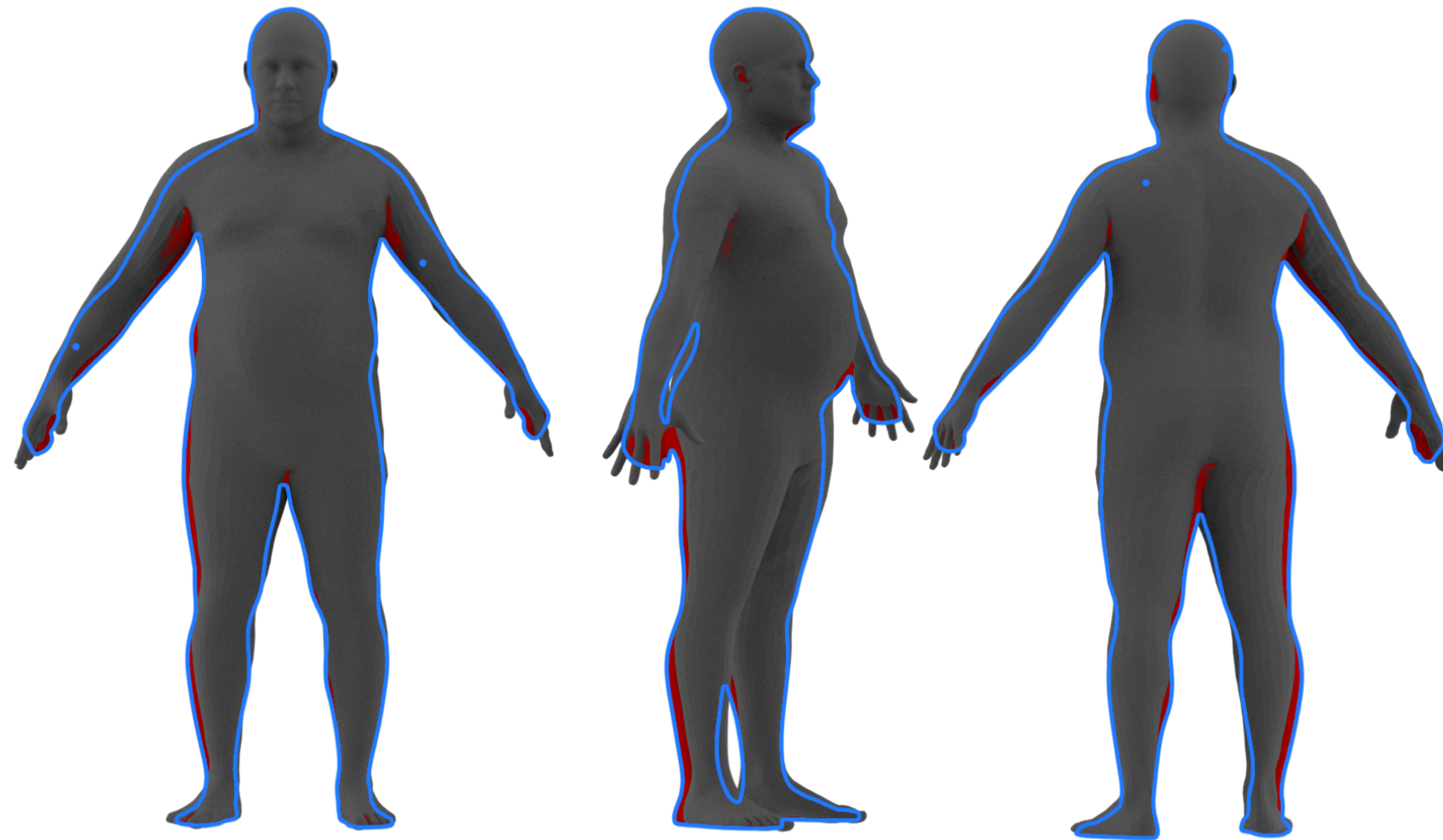


3DLOOK™



As you can see from the comparison, the geometrical error (e.g. point-to-point, surface-to-surface distances) is quite small. The most significant error is located in the waist area and we know that breathing can significantly affect this parameter.

Our 3D models are compatible with all 3D Garment Design software – CLO, Browzwear, Optitex, etc



■ 3D SCAN ■ 3DLOOK



THANK YOU!